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- **Problem Statement ID – SIH26084**
- **Problem Statement Title- Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-125242**
- **Team Name (Registered on portal)- Tech Yodha**



Tech Youdha

Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)

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❖ Challenges and Problems

- ➢ **Coarse Resolution:** Existing NWP models (9–12 km) miss hyper-local convective events (1–3 km).
- ➢ **Fragmented Data:** Reliance on single-source data (only radar or only satellite) fails to capture rapid storm intensification.
- ➢ **Terrain Blind Spots:** Complex mountain terrain blocks radar and creates unpredictable orographic lift, leading to false alarms.
- ➢ **False Alarm Fatigue:** High false alarm rates in mountainous regions cause communities to ignore real warnings.

❖Proposed Solution

➢Rapid-Change Fusion Index (RCFI):
Combines cloud cooling, radar VIL, and lightning jumps into a weighted score. Alerts trigger when RCFI crosses a terrain-adjusted threshold.

➢Arrival Time with Uncertainty:
Provides arrival time with error bounds using ensemble spread.

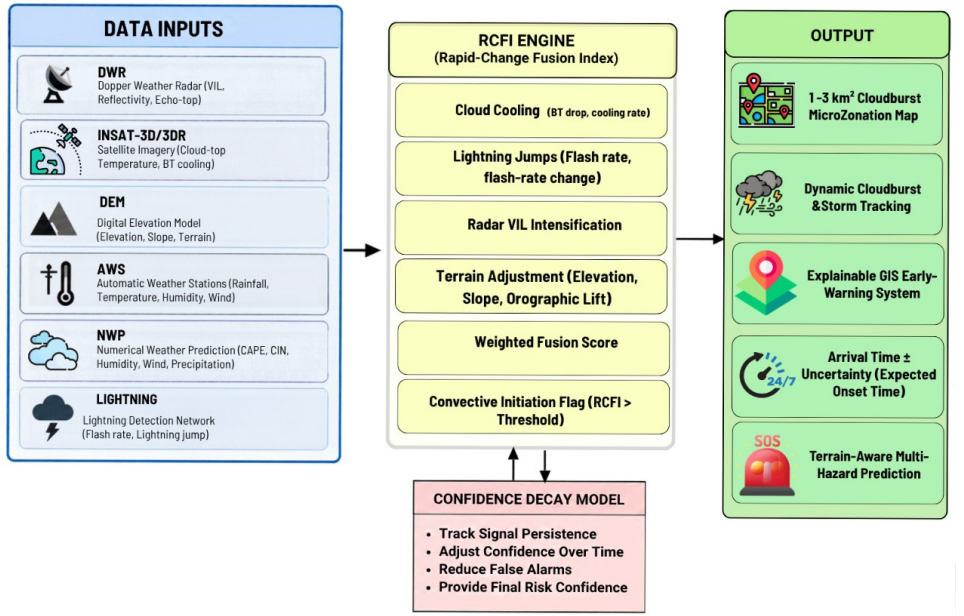
➢Cloudburst Micro-Zonation + Explainable GIS:
Produces a hyper-local cloudburst risk map (0–6 hrs) and shows feature contributions so forecasters can verify the warning

➢Confidence Decay Model:
Prediction trust drops as data latency and lead time increase; stale inputs are automatically downweighted.

❖ Innovation and Uniqueness

➢RCFI Engine: Unified weighted score combining cloud cooling, radar VIL, and lightning jumps.
➢Terrain-Adjusted Thresholds: Warning triggers dynamically adapt to slope, wind, and historical vulnerability.
➢1 km² Micro-Zonation: Hyper-local cloudburst risk grid, not just broad regional maps.
➢Confidence Decay Model: Automatically downweights stale data to reduce false alarms.
➢Explainable GIS: Show exactly why an alert was issued.

❖ How it Addresses the Problems



Tech Youdha

TECHNICAL APPROACH

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❖Methodology and Process

Multi-Source Ingestion: Fuse DWR,Satellite, NWP, and DEM for real-time monitoring.

Preprocessing: Spatial alignment, missing-data handling, and rapid-change feature extraction.

AI Fusion Engine: Spatio-temporal attention models predict cloudburst, lightning, and hail (15 min – 6 hrs).

Dynamic Forecasting: 5–10 min updates on trajectory, arrival time (± uncertainty), and confidence decay.

Explainable GIS: Interactive maps with risk zones, countdowns, and feature attribution.

Decision Support: Location-specific alerts via SMS, dashboard, and API.

❖Technology Stack

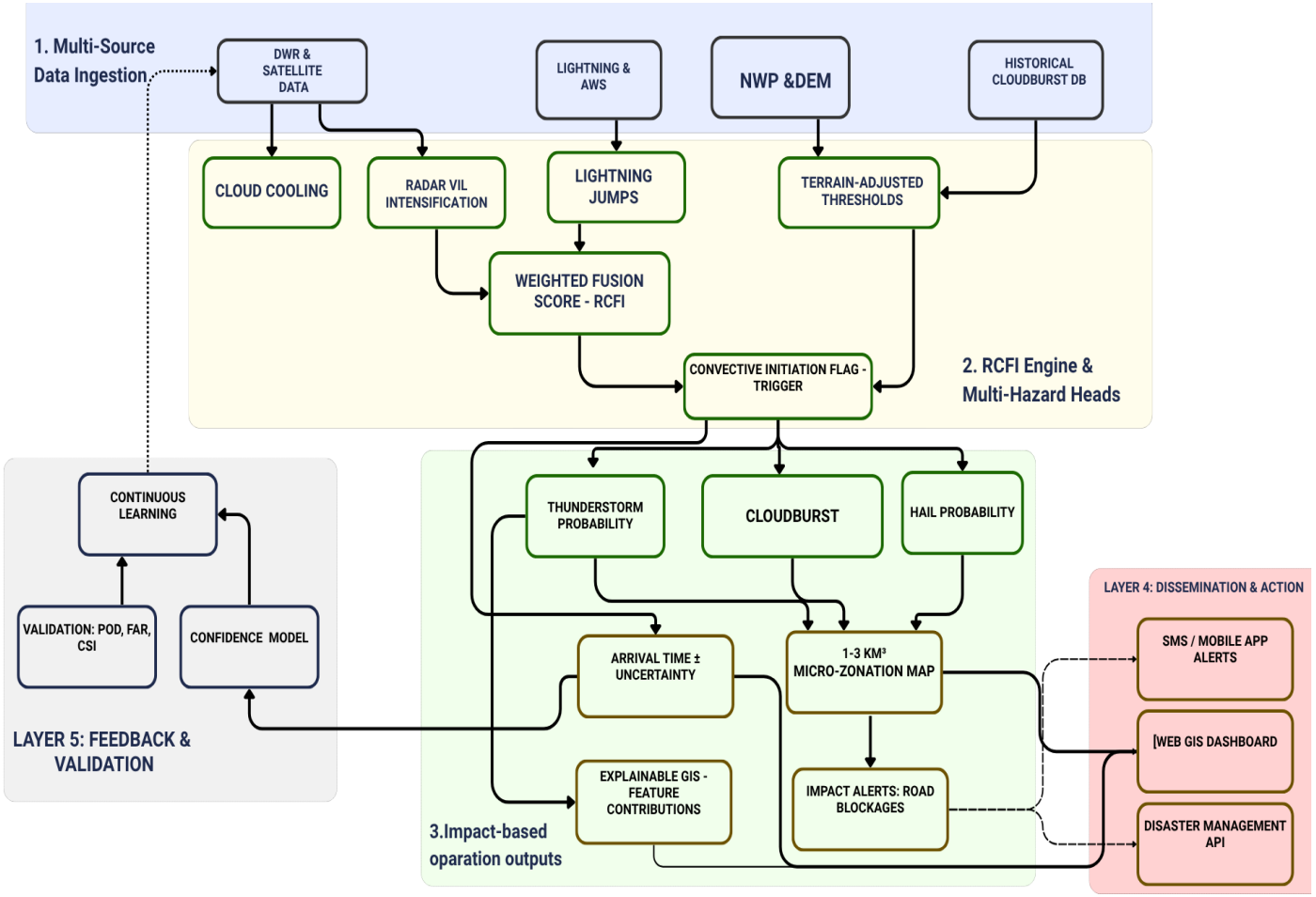
AI/ML: Python, TensorFlow, PyTorch, Scikit-learn,Cv,Pandas

Backend: FastAPI , PostgreSQL (PostGIS)

Frontend & GIS: React,JS,Qgis

Cloud & Deployment: AWS , Docker

❖FLOW CHART



🔗 Project Links:
GitHub:
RCFI Engine – Hyper-local Cloudburst Nowcasting

Demo Video:

Project Progress:
40% Completed.

Your Team Name

FEASIBILITY AND VIABILITY

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IMPACT AND BENEFITS

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RESEARCH AND REFERENCES

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- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

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